Chapter 1
Introduction

The complex, 3D form of the landscape—the morphology of the terrain, the structure of vegetation, and built form—is shaped by processes like anthropogenic development, erosion by wind and water, gravitational forces, fire, solar irradiation, or the spread of disease. In the spatial sciences geographic information systems (GIS) are used to computationally model, simulate, and analyze these processes and their impact on the landscape. Similarly in the design professions GIS and computer aided design (CAD) programs are used to help study, re-envision, and reshape the built environment. However, because of the unintuitive nature of understanding and manipulating 3D forms in abstract, digital space via a graphical user interface (GUI), technologies like GIS are often so challenging to learn and use that they restrict creativity. The limitations of the technology constrain how we think. Being able to interact more naturally with digital space enhances our spatial thinking, encouraging creativity, analytical exploration, and learning. This is critical for designers as they need to intuitively understand and manipulate information in iterative, experimental processes of creation. With tangible user interfaces (TUIs) like Tangible Landscape one can work intuitively by hand with all the benefits of computer modeling and analytics. This chapter discusses the evolution of TUIs and the development of Tangible Landscape, exploring its motivations, applications, and future directions. This chapter also describes the organization of this book.

1.1 The Evolution of Tangible User Interfaces

Inspired by prototypes like Durrell Bishop's Marble Answering Machine (Poynor 1995) and concepts like Fitzmaurice et al.'s Graspable User Interface (1995) Ishii and Ullmer (1997) proposed a new paradigm of human-computer interaction—the TUI. They envisioned that TUIs could make computing more natural and intuitive by coupling digital bits with physical objects as Tangible Bits. In their

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vision Tangible Bits bridge the physical and digital, affording more manual dexterity and kinaesthetic intelligence and situating computing in physical space and social context (Dourish 2001; Ishii and Ullmer 1997). Recently, the development of TUIs has gained momentum thanks to new developments in 3D technologies such as 3D scanning and 3D printing.

1.1.1 The Principles of Tangible User Interfaces

We can easily, intuitively understand and manipulate space physically, but our understanding is largely qualitative. We can also precisely and quantitatively model and analyze space computationally, but this tends to be less intuitive and requires more experience. Intuition allows us to perceive, think, and act in rapid succession; it allows us to creatively brainstorm and express new ideas. TUIs like Tangible Landscape aim to make the use of computers more intuitive combining the advantages of physicality and computation.

Physical models are used to represent landscapes intuitively. With a physical model we can not only see its volume and depth just as we would perceive space in a real landscape, but also feel it by running our hands over the modeled terrain. We can shape physical models intuitively—for example we can sculpt landforms by hand, place models of buildings, or draw directly on the terrain. With a physical model, however, we are constrained to a single scale, simple measurements, and largely qualitative impressions.

Digital models are used to visualize, quantitatively analyze, and precisely design landscapes. With digital models we can dynamically visualize different scales and layers of information, study the landscape by computing geospatial analyses and simulations, and precisely edit digital models. Digital models, however, can be unintuitive. A high level of abstraction separates us from a digital model—to interact with this digital abstraction of physical space with a GUI we must translate our intent through our hands to the computer via input devices like a keyboard and mouse and then visualize the results graphically on a display. The digital model itself is a high level abstraction of physical space represented by arrays of values or by mathematical functions. We have to think abstractly and make great conceptual leaps to work through so many layers of abstraction. Understanding and working directly with a digital model is a hard learned skill, a skill that constrains our thinking and transforms creativity.

Tangible user interfaces aim to make interaction between humans and computers more natural and intuitive by giving digital data a physical form and physical form a digital dimension (Fig. 1.1). By coupling the physical and the digital through a feedback loop, we can interact with our physical data directly and very naturally with our bodies—drawing on our highly developed kinaesthetic intelligence.

A landscape modeled as a physical-digital hybrid through a TUI such as Tangible Landscape has the potential to be intuitively modeled, creatively explored, dynamically computed, richly layered with data, and rigorously analyzed. It can have

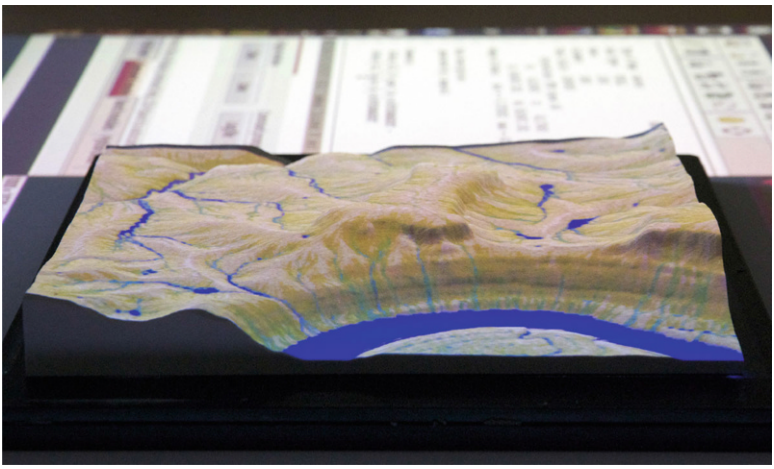

Fig. 1.1 Tangible Landscape: simulated water flow projected over a physical model of a landscape

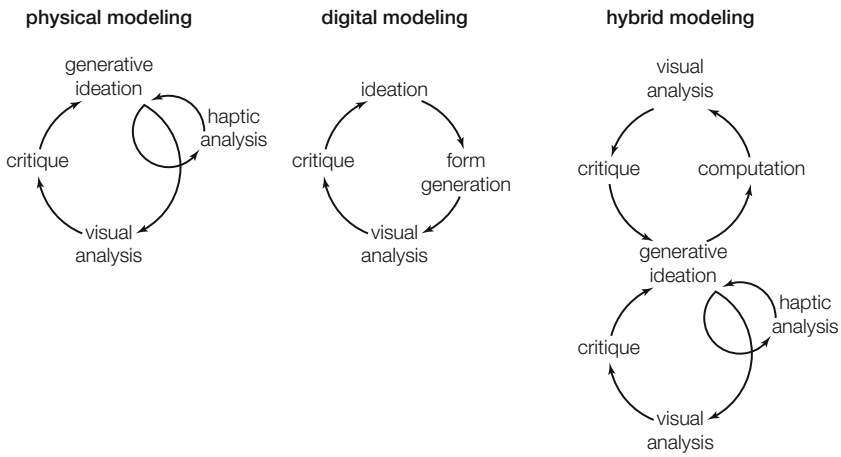

Fig. 1.2 Physical, digital, and hybrid physical-digital modeling processes

all of the advantages of physical landscape models, geospatial models, and CAD combined. With Tangible Landscape we can shape a hybrid model naturally with our hands to creatively explore ideas as if sketching in multiple dimensions. With a physical model we are constrained to modeling in three-dimensions, to modeling for example topographic form. With a hybrid landscape model we are not limited to modeling topographic form—we can dynamically explore and compute many dimensions of geospatial data in rapid succession with the same model (Fig. 1.2).

1.1.2 Dynamic Shape Displays

Dynamic shape displays or shape changing displays—another type of TUI—are computer controlled, interactive, physically responsive surfaces. As we interact with the physical surface it changes the digital model and, conversely, as we interact with the digital model the physical surface changes (Ishii 2008a; Poupyrev et al. 2007). Dynamic shape displays tend to be arrays of pistons and actuated pins that form kinetic, 2.5D surfaces (Petrie 2006) although there is experimental research into continuous, moving surfaces made of shape changing materials driven by heat, magnetic, or electrical stimuli (Coelho and Zigelbaum 2010).

Aegis Hyposurface The Aegis Hyposurface, an early example of a dynamic shape display, is a generative art installation that uses pneumatic actuators to move a triangulated mesh surface according to an algorithm. It can be either preprogrammed or interactive, moving in response to sensed sound, light, or movement. As it was designed and built at an architectural scale the Aegis Hyposurface has a very coarse resolution for an actuated shape display (Goulthorpe 2000).

FEELEX The resolution of actuated shape displays is constrained by the size and arrangement of the piston motors and piston rods or pins that move the surface. Project FEELEX, another early dynamic shape display, used linear actuators to deform a rubber plate. The size of the motors—4 cm—meant that the resolution of the shape display was very coarse. Since the motors are larger than the pins, FEELEX 2 used a piston-crank mechanism to achieve a relatively high 8 mm resolution by clustering the pins while offsetting the motors below. A rubber sheet was stretched over the array of pins to create a 2.5D display for projection. When a user touched the surface they would depress the pins and the pressure of their touch would be recorded as a user interaction (Iwata et al. 2001).

Dynamic Sand Table and Terrain Table The XenoVision Mark III Dynamic Sand Table, developed in 2004, and the Northrop Grumman Terrain Table, developed in 2006, were actuated shape displays that represented topography in 2.5D. In the Terrain Table thousands of pins driven by a motor shaped a silicone surface held taut by suction from a vacuum below into a terrain. The Terrain Table recorded touches as user interactions such as panning and zooming. As users panned, zoomed, or loaded new geographic data, the actuated surface would automatically reshape within seconds (Petrie 2006).

Relief Relief is a relatively low-cost, scalable, 2.5D actuated shape display based on open source hardware and software. Given the complexity and thus the cost, maintenance, and unadaptability of earlier dynamic shape displays like FEELEX and the Northrop Grumman Terrain Table, Leithinger and Ishii (2010) aimed to design a simpler, faster system that was easier to build, adapt, scale, and maintain. In the first prototype of Relief an array of 120 actuated pins driven by electric slide potentiometers stretch a Lycra sheet into a shape display. Users can reshape the shape display by pressing or pulling on the actuated pins. The actuators are

controlled with open source Arduino boards and a program written in the open source language Processing controls, senses, and tracks all of the pins and their relation to the digital model (Leithinger et al. 2009; Leithinger and Ishii 2010). The transparency and freedom of open source solutions should make it relatively easy to reconfigure and adapt this system.

Recompose While Relief was initially designed for a simple, highly intuitive interaction—direct physical manipulation (Leithinger and Ishii 2010)—its next iteration, Recompose, added gesture recognition (Blackshaw et al. 2011; Leithinger et al. 2011). While with Relief users can only directly sculpt the shape display with their hands, with Recompose they can also use gestures to select, translate, rotate, and scale regions of the display. The size and coarse resolution of the actuated shape display mean that only small datasets or subsets of larger datasets can be modeled with useful fidelity. Furthermore, Leithinger et al. (2011) found that only a very limited range of touch interactions could be recognized at the same time and that it can be challenging to manipulate individual pins as they may be out of reach. They augment touch with gestures by adding a Kinect as a depth camera so that users can easily change the context and explore larger datasets. While gestures are less precise than direct physical manipulation, they greatly expand the scope of possible interactions (Blackshaw et al. 2011). Interactions via external devices such as a mouse may be less ambiguous than gestures, but Leithinger et al. argue that they draw users' focus away from the shape display. Therefore they choose to combine touch interactions with gestures rather than pointing devices so that the transition from sculpting to selection, translation, rotation, and scaling would be fluid and seamless given the physical directness of both modes of interaction (Leithinger et al. 2011).

Tangible CityScape Tangible CityScape, a system built upon Recompose, is an example of how this type of TUI can be applied to a specific domain—urban planning. It used a 2.5D dynamic shape display to model and study urban massing. Building masses were modeled by clusters of pins and the model dynamically reshaped as users panned or zoomed with gestures (Tang et al. 2013).

inFORM With inFORM Follmer et al. (2013) developed a dynamically changing user interface capable of diverse, rich user interactions. Building on the Relief and Recompose systems, they developed a 2.5D actuated shape display that supports object tracking, visualization via projection, and both direct and indirect physical manipulation. The shape display is moved by a dense array of pins linked by connecting rods to a larger array of actuators below (Fig. 1.3). The pins, pushing and pulling with variable pressure, offer nuanced haptic feedback to users. A Kinect depth sensor is used to track objects and users' hands. The actuated surface—a grid of pins—can be manipulated directly by pushing and pulling pins. Furthermore, users can interact with the system indirectly via object tracking. The surface can respond to interactions—both direct and indirect—and reshape itself. The actuated surface can also move objects placed on it, enabling indirect physical manipulations. Follmer et al. used this system to explore how shape-changing displays can

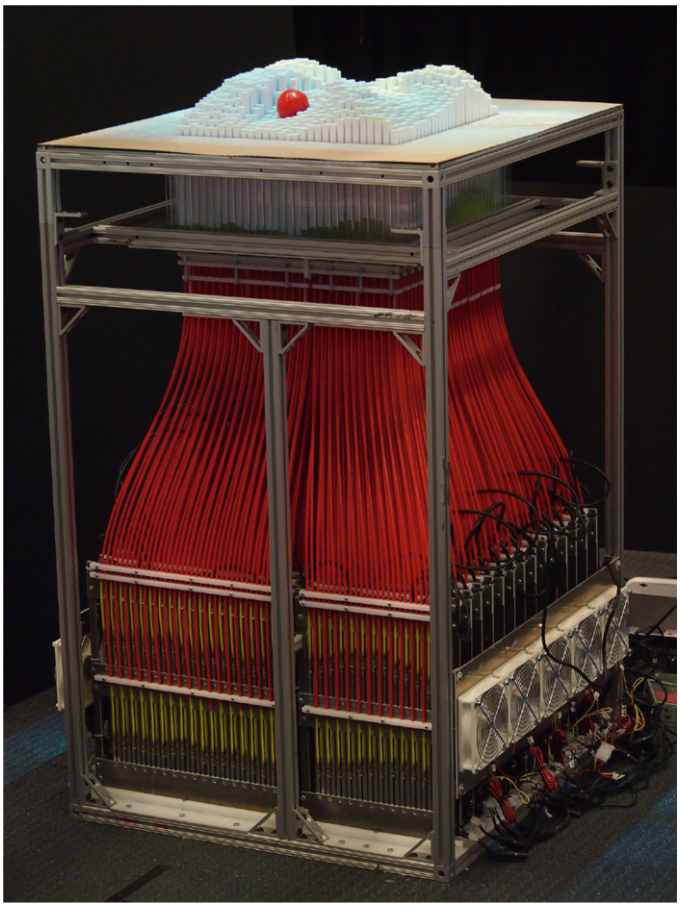

Fig. 1.3 inFORM: a 2.5D actuated shape display designed by the MIT Media Lab (image source: http://tangible.media.mit.edu/project/inform/)

dynamically model content and offer novel modes of interaction based on dynamically changing constraints and opportunities. As a responsive physical user interface inFORM enables rich and varied mode of interaction such as responsive sculpting, moving passive objects, painting changes, and physically instantiated UI elements like buttons (Follmer et al. 2013).

Dynamic shape displays that can be shaped both by touch and computation enable physical interactions from human to computer and computer to human. With a dynamic shape display the digital model is physically instantiated and this shape changing physical model is used for both input and output. When a user changes the surface of the shape display the computer reads the changes; when the computer changes the surface the user sees and feels the changes. Hypothetically this should radically reduce the level of abstraction in human-computer interaction. This novel

mode of bidirectional, tangible interaction should be highly intuitive because it is so direct—human and computers communicate via the same medium. However, while dynamic shape displays may be highly intuitive, they have a relatively coarse resolution, are expensive, maintenance intensive, and hard to transport, all due to the actuators. The resolution of the display for example is constrained by the size of the actuators that make the display kinetic. The coarse resolution makes these displays' representations approximate and abstract, limiting their possible applications.

1.1.3 Continuous Shape Displays

Continuous shape displays couple a continuous physical model with a digital model through a cycle of sculpting, 3D scanning, computation, and projection. We can sculpt the display and then see the resulting computation projected onto the model in near real-time. The feedback from digital modeling is visual—unlike dynamic shape displays, these shape display are passive and do not physically respond to computations. Because there are no actuators the model can be built of a malleable, continuous material such as sand or clay that can be easily sculpted and can accurately represent forms such as landscapes (Ishii et al. 2004; Ishii 2008b).

Projection Augmented Relief Models Currently continuous shape displays rely on projection for representing digital data. While projected imagery has long been used to augment physical terrain models the approach has recently been dubbed Projection Augmented Relief Models (Priestnall et al. 2012). This approach enhances a static physical representation of a landscape with a dynamic visual representation of digital data.

Illuminating Clay With Illuminating Clay (Piper et al. 2002a)—an early continuous shape display from the MIT Media Lab—a clay model of a landscape was continuously scanned with a laser to generate a point cloud of x, y, and z coordinates which were then binned into a digital elevation model (DEM). The DEM or a derived topographic parameter such as slope, aspect, or cast shadow was then projected back onto the clay model so that users could see the impact of their changes in near real-time. Illuminating Clay's library of geospatial analyses was limited as it used custom implementations of elevation, aspect, slope, cast shadow, profile, curvature, viewshed, solar irradiation, and drain direction algorithms. Because Illuminating Clay used a laser scanner the scans were relatively fast—1.2 s each—and accurate to less than 1 mm, but the system was very expensive (Piper et al. 2002a,b). Piper et al. (2002a) also tried to use TUI elements based on object tracking to select algorithms and parameters, but found that too many parameters were involved in geospatial modeling for this approach to be feasible, so they ultimately used standard graphical user interactions with a mouse and keyboard.

Sandscape In an effort to make a more affordable continuous shape display Ishii et al. (2004) designed a system called Sandscape based on infrared depth sensing. In Sandscape a 'sandbox' filled with 1 mm glass beads was lit from beneath by infrared LEDs; an infrared camera above the 'sandbox' captured the intensity of infrared light passing through the glass beads. A DEM based on the light intensity was projected back onto the 'sandbox' so that users would have real-time feedback as they sculpted. While this system was very fast and much less expensive than Illuminating Clay, the scans were low resolution and the physical medium—glass beads—could only be coarsely sculpted (Ishii et al. 2004; Ishii 2008b).

Augmented Reality Sandbox SandyStation (Altman and Eckstein 2014), developed in 2011, and the Augmented Reality Sandbox (Kreylos 2015), developed in 2012, are continuous shape display sandboxes. Kreylos's Augmented Reality Sandbox, inspired by the Czech SandyStation, couples a sandbox with a DEM through a real-time cycle of 3D scanning, modeling and simulation, and projection. As users sculpt the sand a Kinect sensor continually scans the sand surface generating a stream of depth maps. The depth stream is filtered to remove noise and the user's hands. A DEM, contours, and simulated water flow are computed from the depth and projected back on the sandbox. The water flow simulation is a custom implementation based of the shallow water flow equations. The Augmented Reality Sandbox is powered by Kreylos's Kinect 3D Video Capture Project and Vrui VR Toolkit, open source software released under the GNU General Public License. The system is highly intuitive and very fast with each cycle of scanning, processing, and simulation completing in just a couple seconds. However, given that the system only models elevation, contours, and water flow on an abstract landscape it is limited in its scope of potential applications.

Tangible Geospatial Modeling System The Tangible Geospatial Modeling System (TanGeoMS) extended the concept of Illuminating Clay by integrating GRASS GIS (Fig. 1.4). Integrating a full GIS into a continuous shape display gave users a wide range of sophisticated, scientific tools for geospatial modeling and visualization in a highly intuitive environment. By combining freeform hand modeling with geospatial modeling, simulation, and visualization users could easily explore how changes in topographic form affect landscape processes (Mitasova et al. 2006). Like Illuminating Clay the system used a laser scanner to 3D scan a clay model. The scanned point cloud was then interpolated as a DEM, select geospatial analyses or simulations were computed from the DEM, and the results were projected back onto the clay model (Tateosian et al. 2010) (Fig. 1.4b).

1.2 Tangible Landscape

The next generation of TanGeoMS—Tangible Landscape—was designed to intuitively 3D sketch landscapes. This continuous shape display is more tightly integrated with GRASS GIS and has new modes of interaction based on object recognition so that users cannot only sculpt, but also sketch in 3D—rapidly

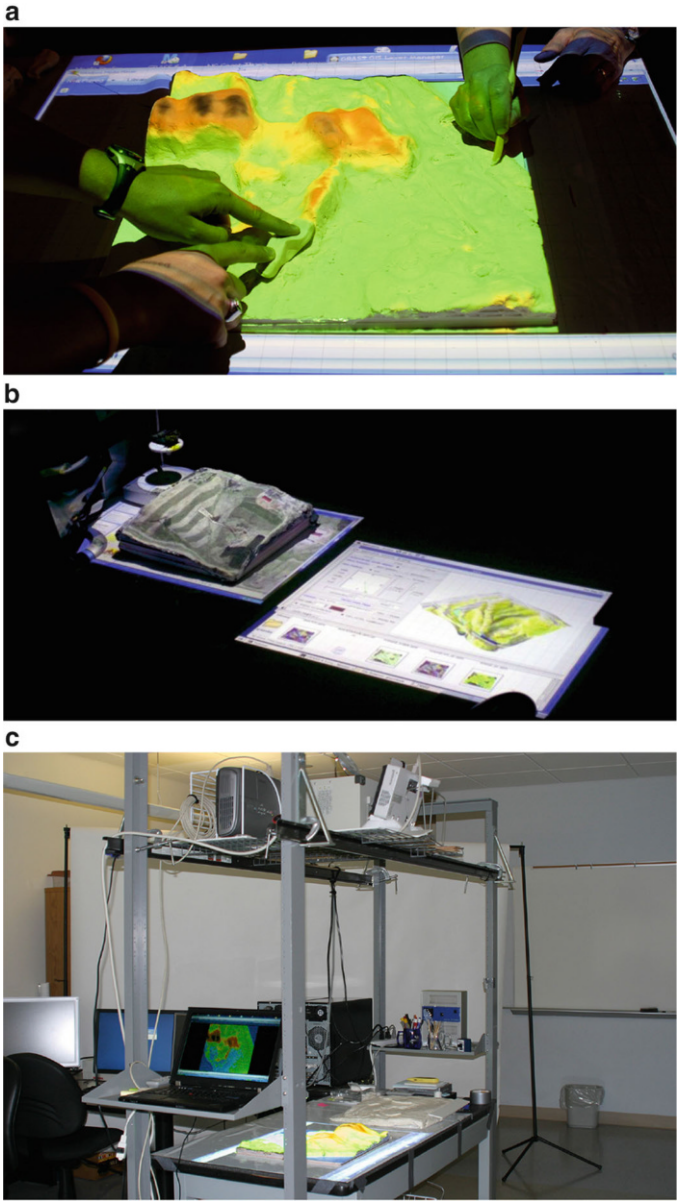

Fig. 1.4 The Tangible Geospatial Modeling System: a continuous shape display powered by GRASS GIS. (**a**) Multiple users interacting. (**b**) The shape display paired with a 3D rendering. (**c**) The hardware setup with a laser scanner mounted above the model

exploring ideas while learning from near real-time feedback (Fig. 1.5). Tangible Landscape is more affordable than its predecessors because it uses a Kinect sensor to 3D scan physical models. As a continuous shape display with object recognition

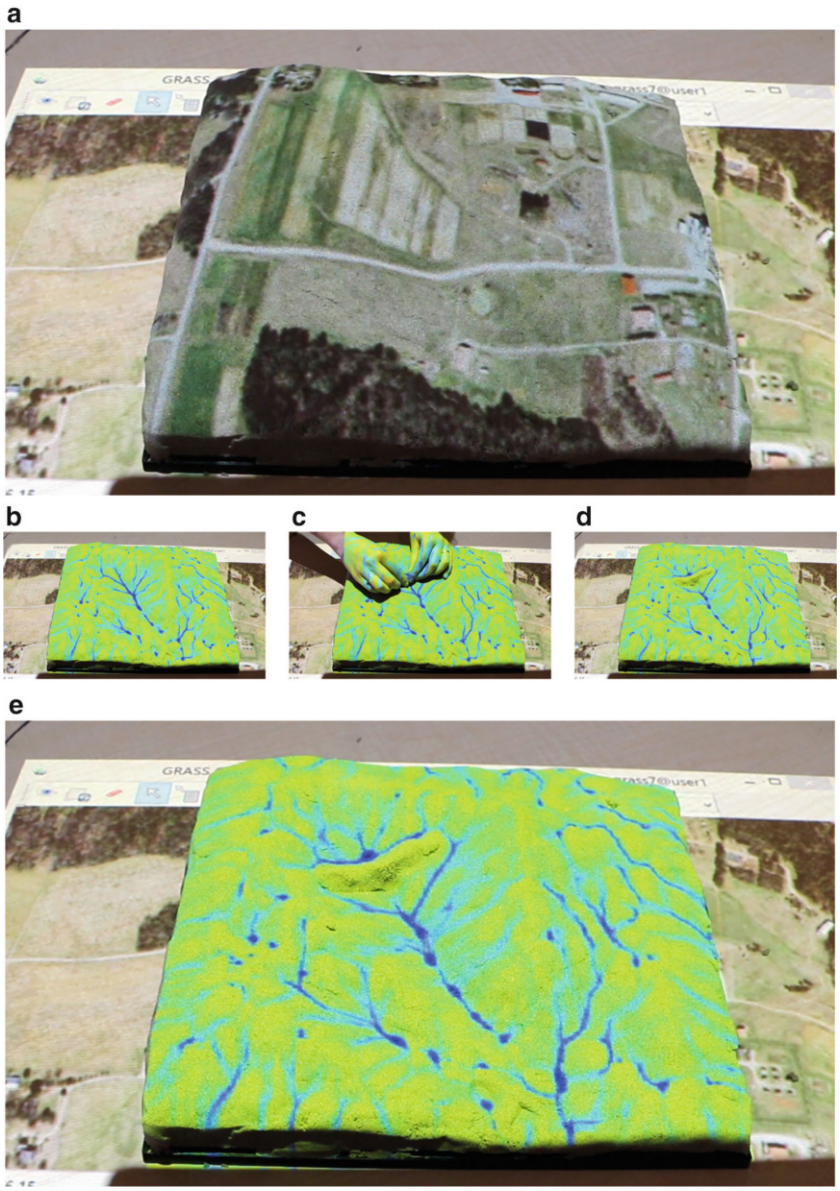

Fig. 1.5 3D sketching a check dam with Tangible Landscape: (**a**) orthoimagery, (**b**) water flow, (**c**) sculpting a check dam, (**d**) updated water flow, and (**e**) a dammed valley with water flow simulated by *r.sim.water*. http://grass.osgeo.org/grass70/manuals/r.sim.water.html

a wide range of media can be used such as polymer-enriched sand, clay, 3D prints, CNC-machined models, and architectural models. It is more tightly integrated with GIS, using a GRASS GIS plugin to automatically scan, process, georeference,

Fig. 1.6 Collaboratively sculpting a retention pond

import, and analyze the model. Because it is so tightly integrated with GRASS GIS users can also use the GUI, the command line, and scripting as advanced controls for tasks not suited to a TUI (Petrasova et al. 2014).

Conceptually, Tangible Landscape couples a physical model with a digital model in a real-time feedback cycle of 3D scanning, geospatial modeling and simulation, and projection. For example, by sculpting the terrain of the physical model, we can see how the changes affect processes like the flow of water, flooding, erosion, and solar irradiation. Thus we can easily and rapidly test ideas while being guided by scientific feedback, exploring a much a larger solution space and make more creative and informed decisions. And since many users can interact with the physical model at once, Tangible Landscape encourages collaboration and interdisciplinary exchange (Fig. 1.6). Tangible Landscape combines real-time interaction with extensive scientific tools for modeling, analysis, and visualization at the precision needed for real-world design and planning applications. Tangible Landscape can be used for applications such as grading landforms (Chap. 3), analyzing topography (Chap. 4), modeling water flow and soil erosion (Chap. 5), analyzing viewsheds (Chap. 6), planning trail networks (Chap. 7), analyzing solar radiation dynamics (Chap. 8), simulating and managing fire (Chap. 9), and modeling inundation and flooding (Chap. 10).

Tangible Landscape is inexpensive, portable, adaptable, and easy to implement. As a high resolution continuous shape display it can be used for both rapid ideation and precise modeling. However, unlike a dynamic shape display the physical model is passive and cannot be changed computationally in real-time—trading real-time response for precision, complexity, and cost. Rapid prototyping can be used to precisely shape or reshape the physical model allowing users to work with real world landscapes and easily reset the conditions.

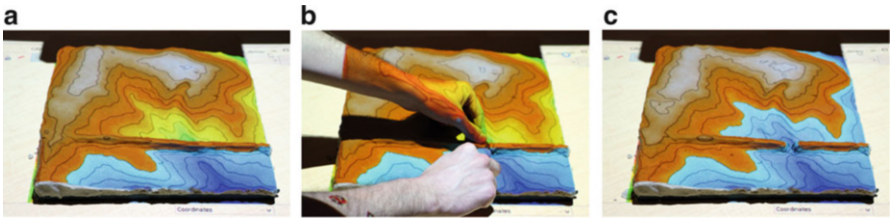

Fig. 1.7 3D sketching a levee breach with feedback on the resulting extent of flooding: (**a**) a flooded landscape, (**b**) breaching the levee, and (**c**) updated flooding

1.2.1 Modes of Interaction

There are many ways to 3D sketch with Tangible Landscape. It combines a couple modes of tangible user interaction. As a continuous shape display—a 2.5D interface that physically instantiates and displays continuous data in real-time—we can use it sculpt surfaces and volumes. It can also use object recognition to detect and compare changes, model volumes, designate points, select data, and sketch lines. By sculpting a physical model made of polymeric sand we can intuitively model topography (Fig. 1.7), forest canopy, buildings, and cost surfaces. We can also explore volumes such as subsurface soil moisture by digging the into polymeric sand. With change detection we can visualize the difference between models to study how we have changed it. With object recognition we can place makers to create points such as observer points for viewshed analysis, lines such as section cuts for visualizing volumes, and polylines such as the least cost path between points for route analysis. Since Tangible Landscape has so many modes of interaction and draws on GRASS GIS' extensive modules for data management, geospatial modeling, and visualization, it can easily be adapted to new uses. Since both Tangible Landscape and GRASS GIS are open source projects we can easily develop new modes of interaction and new analyses.

1.2.2 Developing Tangible Landscape

Tangible Landscape is a free, open source project with source code hosted on GitHub.[1] The system is constantly evolving in response to new developments in its hardware and software components and the needs of its expanding range of applications. Enhanced point cloud processing, optimization of the core functions, and migration to more efficient hardware and operating systems will improve real-time interaction and allow us to incorporate more sophisticated dynamic

[1] https://github.com/ncsu-osgeorel/grass-tangible-landscape

modeling. User interaction and experience can be further improved with dashboards and steered simulations. Computationally controlled shape generation with in-situ digital fabrication will combine the advantages of dynamic and continuous shape displays. By providing Tangible Landscape as a free, open source project, along with this book, we aim to build a community of developers and users of this technology for the benefit of education, research and collaborative decision making in communities worldwide.

1.3 The Organization of This Book

This book describes tangible geospatial modeling using open source GIS and its applications. For each application we explain the underlying theory and algorithms, provide the workflows used, and describe case studies. Simple GRASS GIS workflows consisting of module calls are presented in boxes with green backgrounds, while more complex workflows written in Python are presented in boxes with blue backgrounds. 3D modeling workflows in Rhinoceros are presented in boxes with grey backgrounds.

Chapter 2 describes how Tangible Landscape works and how it is set up. This chapter details the 3D scanning technology, the hardware and software used, and the physical setup of the system. It provides information about where to download GRASS GIS and the Tangible Landscape plugin and explains how to read and use the workflows. Chapter 3 explains how to build physical models for use with Tangible Landscape. This chapter covers handmade modeling techniques, digital fabrication techniques, and techniques for molding and casting models. Chapter 4 covers the basic processes of scanning, interpolating, and analyzing a terrain model. The following chapters describe a range of scientific applications for Tangible Landscape. Each chapter first covers the relevant theory and algorithms and then demonstrates the application in a case study. Chapter 5 covers surface water flow and soil erosion, Chap. 6 covers viewsheds, Chap. 7 covers trail planning, Chap. 8 covers solar irradiation dynamics, Chap. 9 covers the spread of wildfire, and Chap. 10 covers coastal inundation and flooding. The Appendix includes case studies of additional applications of Tangible Landscape, a summary of the data used in this book with download links, a list of useful on-line data sources, and a guide to getting started with GRASS GIS.

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